

THE CELESTIAL SPHERE

ALTHOUGH STARS LIE AT DIFFERENT DISTANCES FROM EARTH, FOR RECORDING THEIR POSITIONS IN THE SKY IT IS HELPFUL TO PRETEND THAT THEY ARE ALL STUCK TO THE INSIDE OF A VAST SPHERE THAT SURROUNDS EARTH.

This enormous imaginary globe is known as the celestial sphere. Every star in the sky other than the Sun, as well as other very remote objects such as galaxies, has a position on the surface of this sphere that remains more or less “fixed”—that is, it hardly changes except over extremely long periods of time. Other, closer objects, such as the Sun and other Solar System bodies, do appear to continuously “wander,” at varying speeds, against the background of stars on the celestial sphere, but they do so in a predictable way.

The sky as a sphere

Like the real sphere of the Earth, the celestial sphere has north and south poles, an equator, and the equivalent of latitude and longitude lines. It is like a celestial version of the globe. The positions of stars and galaxies can be recorded on it as well, just as cities on Earth have their positions of latitude and longitude on a globe. The idea of the sphere also helps astronomers, or indeed anyone, better understand how their location on Earth, the time of night, and the time of year affect what can be viewed in the night sky.

Celestial sphere

The huge sphere on the surface of which stars are imagined to be “fixed”

Ecliptic plane

An imaginary plane on which Earth moves as it orbits the Sun

Orbits of Solar System planets

Most of the other planets orbit the Sun very close to the ecliptic plane

The Sun

The ecliptic

The circle where the ecliptic plane meets the celestial sphere

△ The ecliptic

One of the major circles on the celestial sphere is called the ecliptic. It marks where the ecliptic plane (the plane in which Earth orbits the Sun) meets the sphere's surface. In their motion—as seen from Earth—against the backdrop of stars on the sphere, the Sun always remains on, and planets stay close to, the ecliptic.

Earth's axis of spin
Earth rotates about a line, or axis, that passes through its north and south poles

North celestial pole

A point on the celestial sphere lying directly above Earth's north pole

First point of Aries

One of two points on the celestial sphere where the celestial equator and ecliptic meet

Earth's equator

Ecliptic

The Sun

Unlike other stars, our local star continuously moves on the celestial sphere, although always confined to the line called the ecliptic

Line at right angles to the plane of Earth's orbit around the Sun

△ Imaginary sphere

The celestial sphere is a purely imaginary concept, with a specific shape but no particular size. Astronomers use exactly defined points and curves on its surface as references for describing or determining the positions of stars and various other types of celestial object.

Earth's angle of tilt
The angle, of about 23.4°, between Earth's spin axis and a line at right angles to the plane of its orbit around the Sun

